

Associating geospatially assessed walkability with population health physical activity indicators from urban and regional settings: A case study from Tasmania, Australia

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Submitted: 18 March 2025; Revision requested: 8 October 2025; Accepted: 19 December 2025

Abstract

Introduction: Physical activity is a critical factor in the prevention of chronic disease and in achieving and maintaining good health. However, little is understood about the link between the built environment and physical activity levels in regional areas.

Objectives: We linked geospatially-assessed walkability with population health surveillance data gathered from urban and regional locations across Tasmania, Australia, to assess if walkability was associated with physical activity.

Methods: We used linear regression analysis to determine associations between physical activity duration and the walkability index level generated by geospatial assessments.

Results: We demonstrated a clear association between higher walkability and greater physical activity duration, which was more pronounced in regional areas when compared to urban areas.

Conclusions: Geospatially-assessed walkability may influence physical activity levels in regional areas.

Implications for Public Health: Designing the built environment to increase walkability may be a simple but effective strategy for enabling more physical activity, especially in regional settings.

Key words: walkability, physical activity, public health, regional, quantitative, prevention

Introduction

Regular physical activity (PA) plays a critical and essential role in the prevention of multiple poor health outcomes, including premature mortality, cardiovascular disease, type II diabetes, multiple cancers, depression and anxiety.^{1,2} Furthermore, PA brings substantial social, environmental and economic benefits, both at the individual level and the community level.^{3,4} However, only 24% of Australian adults report meeting the recommended guidelines for PA levels, with regional Australians even less active than their urban adult counterparts.⁵ Residents in regional areas of Australia are also more likely to experience chronic health conditions and engage in risky health behaviours

such as smoking and excessive alcohol consumption, compared with those living in urban settings.⁶

The built environment can positively influence PA through a number of mechanisms, including access to amenities, street connectivity, intersection density, residential density, number of parks, and public transport.^{7,8} However, research on this relationship has largely focused on urban or metropolitan settings,^{9–11} with less known about how the built environment relates to PA in regional settings.¹² Some research indicates differences in the built environment between urban and regional locations,¹³ which may influence the level of health promoting behaviours undertaken by residents, including PA.¹⁴ However, the paucity of evidence related to regional settings

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Aust NZ J Public Health. 2026; Online: <https://doi.org/10.1016/j.anzjph.2026.100316>

provides very limited guidance for policy makers in determining priorities for resource and infrastructure allocation for supportive built environments.

The concept of 'walkability' provides a useful framework for understanding how the built environment can both positively and negatively influence an individual's capacity to undertake PA. Walkability encompasses the concepts of urban design and infrastructure (for example, areas being safe, enticing and traversable), and the outcomes fostered by these environments (for example, encouraging PA and enhancing connectedness through sustainable transport options).¹⁵ Furthermore, a 'walkability index' has previously been established as a method of defining an area's walkability¹⁶ and has previously been positively associated with PA in urban areas,^{17,18} providing a pathway to encourage health-promoting behaviours as part of the built environment.

Research aim

In this study, we linked geospatial assessments of neighbourhood walkability with population health surveillance and monitoring data gathered from both urban and regional suburbs across Tasmania, Australia. From this linkage, we aimed to assess if walkability factors were associated with population-level measures of PA, and if there were differences across urban and regional settings.

Study setting

This study was located in Tasmania, an island state in southern Australia. Tasmania had an estimated population of 571 500¹⁹ in 2022, and is divided into 29 Local Government Areas (LGAs). All regions of Tasmania are classified by the Australian Bureau of Statistics (ABS) as either 'Inner regional', 'Outer regional', 'Remote' or 'Very remote',²⁰ with no regions falling within the 'Major cities' classification. The majority of the population reside in the three major centres of Hobart (population 247 100), Launceston (population 91 000) and the north-west cluster of Burnie, Wynyard, Devonport and Ulverstone (population 74 600). This is supplemented by smaller but economically important regional centres such as St Helens (population 1 573) and Queenstown (population 1 772)²¹ with the remaining 157 000 residents scattered across small towns and settlements (see Figure 1).

Ethics and funding statement

This research was part of the Communities for Walkability study, a project funded by the Australian Medical Research Future Fund (APP1200144) and co-designed with policy partners from the Tasmanian Department of Health (DoH) and the Local Government Association of Tasmania (LGAT). Ethics approval was granted by the Tasmanian Health and Medical Human Research Ethics Committee (ID 23174; 3 March 2021).

Methods

Data sources

Geospatial assessments of walkability

A walkability index was calculated at Mesh Block level for suburbs and towns across Tasmania using standardised scores of local neighbourhood attributes, weighted by the Mesh Block population counts. A Mesh Block is the smallest level of geography included in the Australian Statistical Geography Standard used by the Australian Bureau of Statistics and is similar to a Census Tract. On average, a

Mesh Block contains 30 to 60 dwellings but can be fewer in more remote areas. In Tasmania, Mesh Blocks designated as 'Residential' have an average population of 43.²²

Local neighbourhood attributes included in the walkability index were i) street connectivity (connectivity); ii) dwelling density (density); and iii) a daily living score (destinations), attributes commonly used in walkability and health research.²³ Street connectivity and dwelling density were calculated as the total number of intersections and dwellings respectively within 1600m (1 mile) along the pedestrian-accessible road network from residential addresses. This distance represents a 15-20 minute walk, which is reasonable for regional areas and commonly recognised in the broader literature.^{24,25} These components were calculated at the individual address level before being normalised and added together to create the walkability index. Walkability was then reported at the Mesh Block level by averaging the walkability indexes for each address within the Mesh Block. The third component of daily living score was created using binary indicators recording the presence¹ or absence (0) of access to three types of destinations within 1600m: supermarkets; public transport stops; and convenience stores (including stores, newsagents and petrol stations where basic food is available). These metrics were chosen based on previous research by Gunn et al.²⁶ who found that transport stops, supermarkets and small food or convenience stores are associated with increased walking for transport, and this approach has been used in multiple locations around Australia.²⁷ Distances to these destinations from residential addresses were also calculated using the pedestrian-accessible road network.²⁸ The majority of destinations were sourced through Open Street Map (OSM)^{29,30} and cross-checked with local governments (where available), while road networks were also sourced through OSM. Public transport stops were sourced from General Transit Feed Specification (GTFS) data.³¹ Address points were sourced through the Geocoded National Address File (G-NAF) dataset.³² Population statistics for Mesh Blocks and suburb/town boundaries were sourced from the Australian Bureau of Statistics.²² Results for the local neighbourhood attributes were standardised and aggregated firstly to the Mesh Block and then town/suburb boundaries for further analysis. All analyses were completed in QGIS.³³

Assessment of regionality

Using the Australian Statistical Geography Standard and tables of suburb correspondence²⁰ all suburbs and towns were classified as either 'urban' or 'regional' by assigning 'Urban' to localities classified as 'Inner regional', and 'Regional' to localities classified as 'Outer regional', 'Remote' or 'Very Remote'.

Walkability rankings

The walkability index was first computed for all Tasmanian suburbs and towns and classified into tertiles. As the statewide distribution was heavily skewed towards high scores in urban areas, we standardised the index separately for urban and regional locations to avoid misclassification. Within each stratum (urban or regional), we linearly scaled the index to a 0-1 range using the following formula,

$$stdwalk_{index} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

where x is the raw index. We then redefined walkability levels to Low, Moderate or High using tertiles of the standardised values within each stratum. (see Table 1).

Figure 1: Map of Tasmania, Australia, showing ABS regional classifications and major population centres.

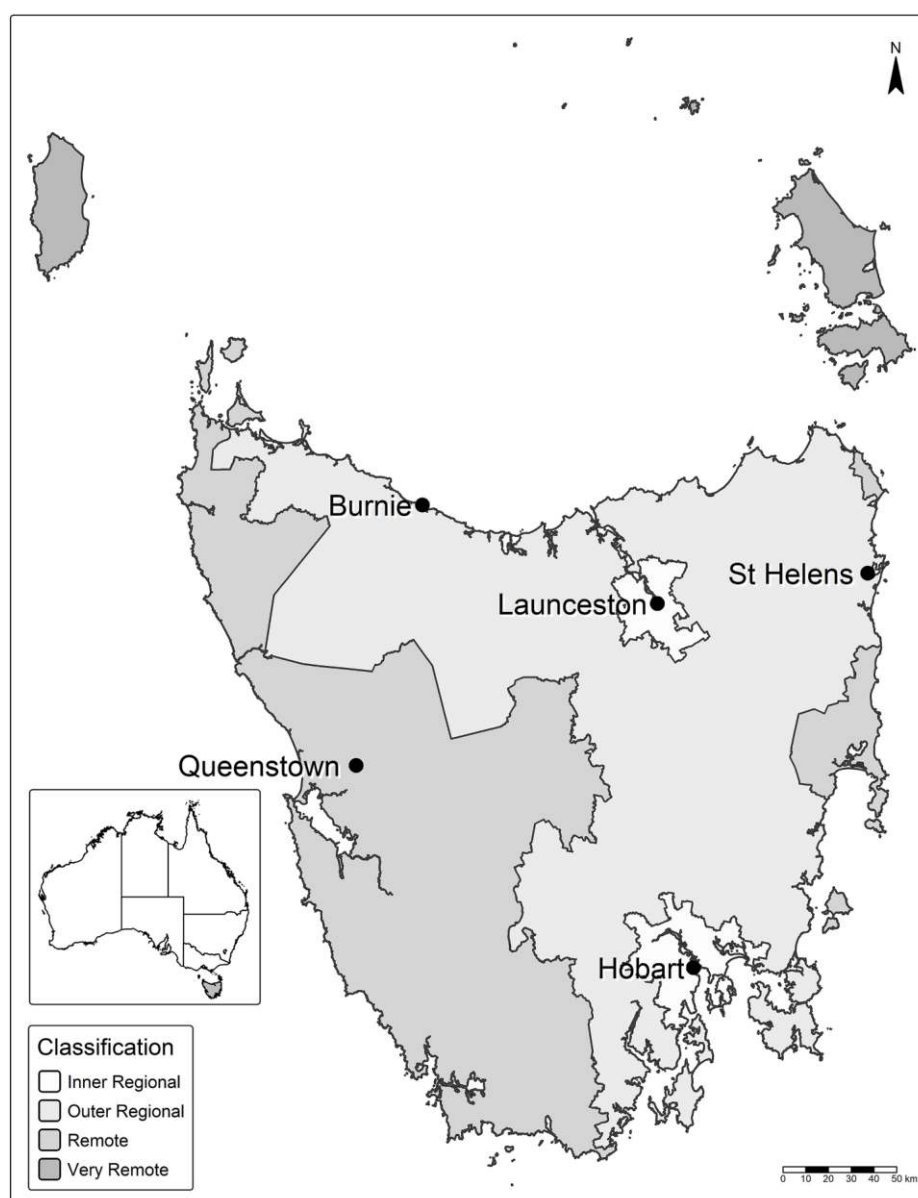


Table 1: Walkability index categories for urban, regional and all towns in Tasmania, before and after normalisation applied.

Walkability category	Before normalisation		After normalisation	
	Urban	Regional	Urban	Regional
High	124	113	77	159
Moderate	25	93	39	80
Low	6	113	39	80

Tasmanian Population Health Survey data

The Tasmanian Population Health Survey (TPHS) is a three- to four-yearly, cross-sectional population health surveillance and monitoring survey of a representative sample of Tasmanians aged 18 years and over, which has been undertaken since 2009. In 2022, recruitment was via the Integrated Public Number Database (IPND), a mobile telephone number database with associated postcode information.

This allowed for targeted recruitment by LGA, with a computer-generated list of random numbers from each LGA. Using standardised protocols, a research company contracted by the DoH conducted telephone interviews and collected information on sociodemographic variables (for example, age, gender and education), health behaviours (for example, PA and diet), and physical and mental health outcomes (for example, body mass index, psychological distress and the presence of various chronic conditions).

Participants were informed about the requirements of voluntary participation (a telephone interview), what the data would be used for (to support planning and research, implementation and evaluation of health services and programs in Tasmania). There were 4,061 participants in total, with a response rate of 29%, calculated as a percentage of completed interviews from all interviews and refusals.³⁴ To ensure adequate representation across Tasmania,

quotas were set for each LGA, although the survey under-represented males and younger age groups when compared to 2021 Tasmanian Census data.³⁴

For this analysis, we evaluated four PA outcomes. These included duration (mins/week) of walking, moderate-intensity PA, vigorous-intensity PA, and moderate-vigorous PA (MVPA, calculated by summing moderate PA and vigorous PA outcomes) (see Supplementary Materials for PA-related survey questions). PA outcomes were aggregated to the town/suburb level, and the mean calculated. Towns or suburbs with no PA data were omitted from analysis. While walking and moderate-intensity PA are conceptually more closely aligned to walkability, we included an exploratory analysis of the vigorous-intensity PA and MVPA outcomes as it is plausible vigorous activity could be influenced by neighbourhood walkability.

Data linkage

TPHS data were provided with approval by the Tasmanian DoH. Linkage of geospatial walkability data to 2022 TPHS data were based on the location (suburb or town) provided by the TPHS respondent. Some suburbs recorded as places of residence by TPHS respondents were recorded as having zero population in the 2021 Census, due to differences in data collection methods between the 2021 ABS Census and the 2022 TPHS. For these respondents, we used Tobler's First Law of Geography³⁵ and reallocated suburb of residence to the nearest town with a non-zero population recorded by the 2021 Census.

Statistical analysis

Sociodemographic data from the TPHS were summarised using proportions for categorical variables. Linear regression analysis was used to determine associations between PA (minutes) and the walkability index. Using the 'low' category as a baseline, comparisons were made between 'low' and 'moderate' walkability categories, and 'low' and 'high' walkability categories. Given the small number of residents in some towns used in the analysis, population weights were also applied. These were derived by calculating the proportion of residents represented from the general population of the town.²²

To address non-normality in the PA outcomes, the Yeo-Johnson transformation³⁶ was applied as it handles positive, zero and negative values without shifting the data. This transformation improves model validity by stabilising variance and enhancing interpretability. Model selection was guided by the Akaike Information Criterion (AIC) comparing weighted versus unweighted models for each outcome. We used weighting rather than covariate adjustment, as the weighted models had better AIC than both the unweighted models and the models including the proportion of residents as a covariate. Regression estimates with 95% confidence intervals (CIs) were reported.

To determine differences in PA duration in minutes, expected marginal means (EMMs) by walkability level were obtained from the transformed-scale linear models and back-transformed using parametric bootstrapping with 5,000 draws. Bootstrap samples from the multivariate normal of the EMMs (using the estimated mean and variance-covariance estimates) were mapped through an inverse Yeo-Johnson function using outcome-specific lambda values estimated from the preprocessing step, yielding EMMs in minutes with 95% percentile CIs. Pairwise differences in minutes (low vs high,

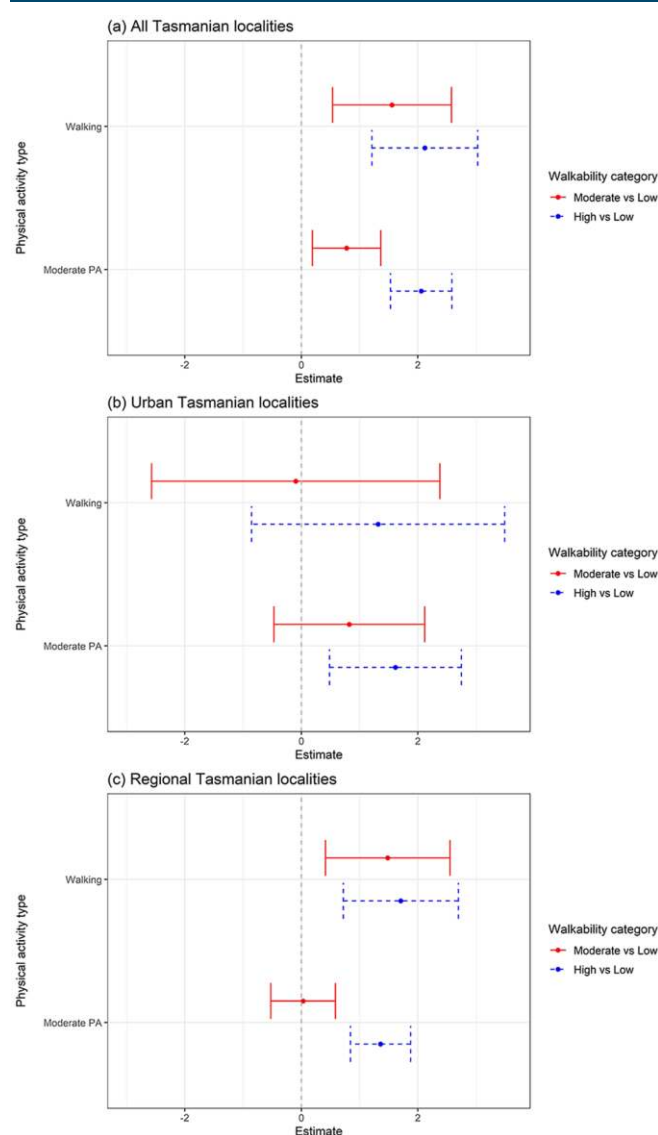
low vs moderate) were computed, with two-sided p-values from the proportion of bootstrap differences crossing zero. All analyses were conducted using the *lm()* function in R.³⁷

Results

Tasmanian Population Health Survey results

The TPHS had 4,014 respondents, with 65.8% (n=2540) living in an urban area, and 34.2% (n=1374) in a regional area. Sociodemographic characteristics of survey respondents can be found in [Supplementary Table 1](#). When compared with the TPHS dataset as a whole, urban locations have a greater proportion of residents who are employed, earn more, are younger and have higher educational attainment, compared to those living in regional locations.

Figure 2: Population-weighted association between physical activity type (minutes per week of walking and moderate PA) and walkability categories in (a) all Tasmanian localities, (b) urban Tasmanian localities, and (c) regional Tasmanian localities.



Suburb/town results

Of the 775 suburbs and towns in Tasmania, 478 were included in this analysis. The remaining towns did not have representation in the TPHS. Suburbs and towns included in analyses had resident populations between 8 and 14,481, with an average population of 4,080. Urban localities had a range of 58 to 14,481 and an average population of 5,339, while regional locations had a range of 8 to 6,653 and an average population of 1,659.

Regression analysis

Figure 2 (and Supplementary Table 2) show the population-weighted association between walking and moderate PA with the walkability index category for all of Tasmania, and separately for urban and regional areas. For unweighted population results, see Supplementary Table 3.

Figure 2a shows the association between walking and moderate-intensity PA with walkability index categories across all of Tasmania. Overall, regions with higher walkability were associated with more minutes of walking and moderate-intensity PA, when compared to regions with lower walkability index scores. There was a stronger association observed when comparing high to low walkability categories than when comparing moderate to low categories, showing a dose-response gradient (i.e. 'high' categories demonstrated greater duration of PA than 'moderate' or 'low' categories).

Figure 2b shows the association between walking and moderate-intensity PA with walkability index categories for urban locations in Tasmania. Similar to results for all of Tasmania, a dose-response gradient was observed for walking and moderate-intensity PA with high versus low walkability categories of walkability, compared to moderate versus low categories. However, the association was only significant for moderate-intensity PA in high versus low walkability regions. Skewed distributions of walkability categories for the urban regions, even after normalisation (see Table 1) may account for greater confidence intervals in these results.

Figure 2c shows the association between walking and moderate-intensity PA with walkability index categories for regional locations in Tasmania. Regions with higher walkability index scores were

Table 2: Population-weighted estimated differences (in mins) in duration of walking and moderate PA for all Tasmanian localities.

PA type	Category	Difference (mins)	p-value	Low CI	High CI
Walking	Low vs Moderate	58.93	<0.05	19.65	100.64
Walking	Low vs High	84.11	<0.05	47.40	122.16
Moderate PA	Low vs Moderate	6.46	<0.05	1.37	12.81
Moderate PA	Low vs High	31.56	<0.05	20.62	45.30

Table 3: Population-weighted estimated differences (in mins) in duration of walking and moderate PA for urban Tasmanian localities.

PA type	Category	Difference (mins)	p-value	Low CI	High CI
Walking	Low vs Moderate	-2.17	0.94	-70.56	64.97
Walking	Low vs High	38.24	0.23	-23.57	100.00
Moderate PA	Low vs Moderate	13.28	0.20	-7.24	35.77
Moderate PA	Low vs High	30.25	<0.05	9.90	51.91

Table 4: Population-weighted estimated differences (in mins) in duration of walking and moderate PA for regional Tasmanian localities.

PA type	Category	Difference (mins)	p-value	Low CI	High CI
Walking	Low vs Moderate	63.91	<0.05	17.41	113.56
Walking	Low vs High	75.00	<0.05	31.17	121.36
Moderate PA	Low vs Moderate	0.23	0.898	-2.77	3.49
Moderate PA	Low vs High	18.13	<0.05	9.37	30.15

associated with greater duration of resident walking, when compared to regions with lower walkability index scores, in a dose-response manner. There was a significant association between moderate-intensity PA for high versus low walkability categories, but no association with moderate versus low walkability categories was observed.

Population-weighted estimated differences in duration of walking and moderate PA are shown in Table 2 (all Tasmanian localities), Table 3 (urban localities) and Table 4 (regional localities). Unweighted estimated differences are presented in Supplementary Materials (see Supplementary Table 4).

Exploratory analyses

Exploratory analyses were performed for vigorous-intensity PA and MVPA, as it is plausible that variables are influenced by neighbourhood walkability. Results for these outcomes are presented in Supplementary Materials (see Supplementary Fig. 1 and Supplementary Table 5). Overall, regions with higher walkability index scores trended towards increased duration of resident vigorous PA and MVPA, when compared to regions with lower walkability. This association was statistically significant when low and high walkability categories were compared, but not when comparing moderate to high walkability categories. For urban locations in Tasmania, associations trended in the opposite direction to expectations but were not statistically significant. For regional locations in Tasmania, associations between vigorous PA and MVPA with moderate versus low walkability categories were in the opposite direction to expectations and statistically significant, while associations between vigorous PA and MVPA with high versus low walkability categories were in the expected direction but did not reach statistical significance. Overall, these disparities indicate no clear pattern between vigorous PA, MVPA and walkability index.

Discussion

Our findings indicate that greater spatially-assessed walkability was associated with a higher duration of both walking and moderate PA in Tasmania. This association was stronger with higher walkability levels and for regional areas compared to urban areas. Across Tasmania, differences in walking duration of close to one hour were shown when comparing low to moderate walkability-assessed areas, and over 80 mins when comparing low to high walkability-assessed areas. Increases of over six minutes to over 30 minutes were demonstrated for moderate PA duration.

This is one of the first studies in adults that we know of examining the association between an objective geospatial walkability index assessment and PA in regional areas of Australia. Our findings are important for regional health outcomes, and suggest that built environments in these areas need to be improved and made more

walkable to increase PA among residents. With regional adults globally demonstrating higher rates of premature mortality and chronic disease,^{38,39} designing and enhancing built environments to support walking and moderate PA could provide a critical avenue for improving the health and wellbeing of these populations.

Across the state as a whole, there was a clear and consistent dose-response association between higher walkability index scores and greater durations of walking and moderate-intensity PA. This is consistent with findings from the existing (mostly urban) literature, which has shown the importance of a range of factors for PA, including proximity of destinations, land use mix, connectivity, population density, neighbourhood aesthetics, safety from crime, and proximity to parks.^{40–42} The objective spatial measures of walkability used in this study were a summary index of connectivity, density and destinations, and designed for use in urban areas. The current results confirm the applicability of these measures at the population level in both urban and regional settings with clear and consistent associations between walkability and PA across the state, even in small towns.

When stratified by urban/regional location, however, findings were less consistent. There was no evidence of an association between walking or moderate-intensity PA in urban areas with moderate compared to low levels of walkability. However, a positive association was apparent between walking and moderate-intensity PA in urban areas with high versus low levels of walkability, albeit statistically significant only for moderate PA. This is possibly a dilution effect, where urban-dwelling residents have more options for spaces and places to be active. For example, when the immediate environment does not cater to their needs, they may be able to more easily travel to other areas to be active, whereas this option may not be available to regional residents.

In regional areas, both moderate and high levels of walkability were positively associated with walking, while high levels of walkability were positively associated with moderate-intensity PA. These findings suggest walkability is especially important for PA in regional communities, particularly for walking. Walking is the most common and freely available form of PA for adults⁴³ and is a key behavioural target for PA promotion, given it is accessible and typically does not require specialised equipment, clothing, or skills. With the high prevalence of premature mortality and chronic disease and lower prevalence of PA in regional areas,^{38,39} creating more walkable built environments holds promise as an important opportunity to promote active lifestyles at the population level. Planning and design policies for regional areas that support street connectivity, have a variety of destinations, and greater population densities should be considered as public health strategies to increase PA and improve health in regional communities.

In our exploratory analyses, we found little evidence of an association between walkability and vigorous-intensity PA or MVPA. At the state level, associations with high compared to low walkability levels were in the expected direction for both vigorous PA and MVPA. However, when stratified by urban/regional area, associations were inconsistent in direction and magnitude across levels of walkability, across PA type (vigorous PA and MVPA), and across areas (urban/regional), suggesting these stratified results should be interpreted with caution, and may be due to the smaller number of participants reporting vigorous PA in the TPHS. Plausibly, the types of vigorous PA performed may be more likely to occur outside of the local

neighbourhood – for example, at sports centres or running/cycling trails and tracks that may be more geographically dispersed – and hence walkability may be less likely to play a strong role in supporting/hindering some vigorous activities.

Policy implications

Across Tasmania, differences in walking and PA duration between categories ranges from 6.5 minutes per week (of moderate PA in low versus moderate walkability towns) to 84.1 minutes per week (of walking in low versus high walkability towns). This equates to 4.3% to 56% of the minimum recommended duration of moderate PA (150 minutes per week), which includes brisk walking.⁴⁴ At the higher end, this is an important and substantial contribution to meeting PA guidelines. While out of scope of this study, significant investment in walkability factors, especially in regional centres, is likely to be worthy of consideration to increase PA participation, which may lead to reductions in chronic health conditions.

Numerous public health policies and frameworks already highlight the importance of walkability and the built environment in supporting communities to be more active. Key policies include the World Health Organization's Global Action Plan for Physical Activity,⁴⁵ Australia's National Preventive Health Strategy,⁴⁶ and the Heart Foundation's Blueprint for an Active Australia.⁴⁷ Design guidelines such as the Heart Foundation's 'Healthy Active by Design' also emphasise the importance of designing communities that promote health.⁴⁸ However, few have a specific focus on understanding walkability and its role in promoting PA in regional areas. The findings from this study suggest that designing walkable communities may be equally, if not more influential on PA in regional areas, and must be given adequate attention in planning and design considerations. All communities should have access to environments that support active living, including those living in regional areas. Addressing current gaps in the walkability of regional areas is needed to increase PA participation and improve health and wellbeing, reducing existing geographic disparities in health.

Local government planning also has a critical role to play to increase walkability and encourage both recreational and transport walking and increased PA in regional locations. Urban design in these areas is usually focused on a main street with key destinations clustered in the centre of town. However, if a regional population grows rapidly, new housing is located on the edge of town in low density developments well beyond distances of within 800m from the key destinations that encourage walking.^{26,49} In Australia, regional migration patterns have changed since the COVID-19 pandemic⁵⁰ and are likely to continue if residents can retain metropolitan labour market employment in nearby larger regional towns,⁵¹ or post-pandemic working from home remains viable.⁵² This requires new responses from local government to review strategic and statutory planning practices, including those focused on density destinations and connectivity to support walkability, PA and communities designed to support health.

Finally, improved walkability may have benefits beyond PA, with the connection between walkability, social connectedness and inclusion an additional and important perspective when considering wider community health outcomes. A positive association between walkability and social connectedness has been repeatedly demonstrated,^{53–55} while social and family connectedness has been

linked to improved mental health, improved relationships, and reduced substance abuse.^{56–58}

Strengths and limitations

There are a number of strengths and limitations of this study that should be considered when interpreting the findings. Firstly, data were cross-sectional and analysed at the level of localities (towns/suburbs). This means there is the potential for ecological fallacy and caution is needed in making inferences about individuals from inferences about the group which those individuals belong. For example, active people could choose to live in more walkable areas, although it has been demonstrated that residential self-selection plays a limited role in built environment/PA associations.⁵⁹ A further limitation is that confounders such as socioeconomic or demographic characteristics were not included in analysis, however population-weighting was considered in the model. The geospatial assessment of walkability is both a strength (i.e. an objective measure reduces the risk of bias) and a weakness (i.e. there is lack of qualitative or lived experience data) but it is a standardised measure (i.e. walkability index) that has been used across Australia in research²⁷ and policy.⁶⁰ The use of population surveillance and monitoring data to estimate PA is also both a strength (i.e. population coverage, random sampling, efficient and economical, maximises an existing resource) and a weakness (i.e. 29% response rate, no control over study design and measures). The use of a self-reported measure of PA is a potential limitation, however, the questions asked were based on a reliable and validated measure recommended for use by all Australian jurisdictions as the tool for monitoring physical activity in adults.⁶¹ Finally, although we refer to 'urban' localities, the entire state of Tasmania is classified by the Australian Bureau of Statistics as regional and remote,²⁰ which may result in a lack of heterogeneity, where the more urbanised areas in this study may differ from urban areas in other states in Australia.

Future research

This study highlights the usefulness of linking geospatial walkability data with routine health surveillance data collected from people in both urban and regional areas to gain greater insights for population health and planning, and indicates that future health surveillance activities could include a geospatial component. While geospatial data has been linked with routine surveillance data in the communicable disease space,⁶² its application in relation to health behaviour surveillance is less common. Doing so would have application beyond PA research – for example, to fast food, alcohol and tobacco retail outlets and their associations with unhealthy diets, risky alcohol consumption and tobacco use. Qualitative research at the individual level would allow greater inferences about associations between the built environment and PA behaviour. Integrating qualitative data into geospatial assessments of walkability is likely to provide greater insights into lived experience and the broader context of behaviours. The use of objective measures of PA would strengthen future research, although at greater cost. Including comparisons with more urbanised areas may provide greater heterogeneity and different understandings of associations. Finally, an exploration of how sociodemographic or socioeconomic differences could account for differences in PA beyond regionality would add to this evidence base, as would a more comprehensive

exploration of the impact of walkability on social connectedness in regional and remote settings.

Conclusion

This research demonstrates the utility of linking geospatially-assessed walkability data with population surveillance survey data in a regional setting. We found greater walking and moderate-intensity PA in areas with higher geospatially-assessed walkability, when compared to areas with lower geospatially-assessed walkability. This trend was stronger in regional areas, providing an overlooked opportunity to support these communities to participate in more PA, meet PA guidelines, and achieve better health outcomes. These findings may assist urban planning and health policymakers to understand the implications of built and landscape design decisions on population health outcomes, especially in regional environments.

Conflicts of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

Funding

This work was financially supported by the Australian Medical Research Future Fund (APP1200144), in partnership with the Tasmanian Department of Health and Local Government Association of Tasmania, Australia Heart Foundation and University of Tasmania, Australia. The funding source had no involvement in the study design, collection, analysis and interpretation of data or manuscript development. VC was funded by a Heart Foundation Future Leader Fellowship (ID 104892).

Data availability statement

Data may be made available on request.

Acknowledgements

The authors gratefully acknowledge Public Health Services (DoH, Tasmania) for provision of TPHS data. The authors also acknowledge Dr Myriam Ziou (University of Tasmania) for additional assistance with statistics, Dr Grant Williamson (University of Tasmania) for additional assistance with mapping, and the numerous developers of R packages used in analysis.

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Appendix A Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.anzjph.2026.100316>.